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# NEET FLIX

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## CHEMISTRY

1) **Statement I :** An atom having three electrons in outermost shell and $ns^2np^6nd^{10}$ configuration in penultimate shell, belongs to p-block **Statement II :** The above element belongs to 3rd period.

(1) Both the statements I and II are correct
(2) Both the statements I and II are incorrect
(3) Statements I is incorrect but Statement II is correct
(4) Statements I is correct but Statement II is incorrect

2)

| **Column-I Element** | | **Column-II Period and group number respectively** | |
|---|---|---|---|
| (P) | Si | (1) | 4, 3 |
| (Q) | Sc | (2) | 3, 14 |
| (R) | Ga | (3) | 6, 13 |
| (S) | T□ | (4) | 4, 13 |

(1) P→2;Q→1;R→3;S→4
(2) P→1;Q→2;R→4;S→3
(3) P→2;Q→1;R→4;S→3
(4) P→4;Q→3;R→1;S→2

3)

Largest halogen ion is :

(1) $H^-$
(2) $F^-$
(3) $I^-$
(4) $Br^-$

4) Given that $Z_{eff.}$ value of 'N' (Z = 7) is 'Y'. Then which of the following statement is correct

(1) σ value for 'N' will be 7 + Y
(2) σ value for oxygen will be 7.35 – Y
(3) σ value for oxygen will be 7 – Y
(4) Given information is not sufficient

5) Which of the following has $Z_{eff}$ equal to Z :

(1) $C^{+2}$

(2) $B^{+3}$

(3) $N^{+3}$

(4) $Li^{+2}$

6)

Which of the following statement(s) is/are correct:-
(a) Vander waal's radius of iodine is more than its covalent radius
(b) All isoelectronic ions belong to the same period of the periodic table
(c) $IE_1$ of N is higher than that of O while $IE_2$ of O is higher than that of N

(d) The 1st electron gain enthalpy of Cl is negative while second is positive.

(1) a,b

(2) a,b,c

(3) a,c,d

(4) a,b,c,d

7) The 1st, 2nd and the 3rd ionization enthalpies $I_1$, $I_2$ and $I_3$, of four atoms with atomic numbers n, n + 1, n + 2 and n + 3, where n < 10, are tabulated below. What is the value of n?

| Atomic Number | Ionization Enthalpy (kJ/mol) | | |
|---|---|---|---|
| | $I_1$ | $I_2$ | $I_3$ |
| n | 1579 | 3374 | 6043 |
| n+1 | 2075 | 3952 | 6125 |
| n+2 | 493 | 4562 | 6907 |
| n+3 | 739 | 1451 | 7731 |

(1) 6

(2) 7

(3) 8

(4) 9

8) The incorrect statement is

(1) The first ionization enthalpy of K is less than that of Na and Li

(2) Kr does not have the lowest first ionization enthalpy in its group

(3) The first ionization enthalpy of element with atomic number 37 is lower than that of the element with atomic number 38.

(4) The first ionization enthalpy of Ga is higher than that of the d-block element with atomic number 30.

9)

| **List-I (Atom / Ions)** | **List-II (Electron Affinity in eV / atom and where EA = $-\Delta H_{e.g.}$)** |
|---|---|

| | | | |
|---|---|---|---|
| (P) | F | (1) | 3.4 |
| (Q) | $F^+$ | (2) | 17.4 |
| (R) | Cl | (3) | 13 |
| (S) | $Cl^+$ | (4) | 3.6 |

(1) P-1; Q-3; R-4; S-2

(2) P-2; Q-1; R-3; S-4

(3) P-1; Q-2; R-4; S-3

(4) P-3; Q-1; R-2; S-4

10)

Endothermic process is :

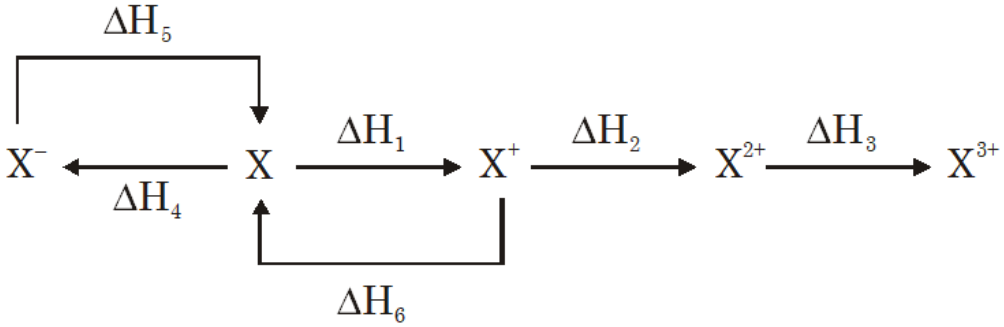
(1) $N \rightarrow N^-$

(2) $P \rightarrow P^-$

(3) $F \rightarrow F^-$

(4) $N^- \rightarrow N$

11) Consider the following sequence of reaction:




If electronic configuration of element X is $[Ne]3s^1$, then which of the following order is incorrect regarding given enthalpies.

(1) $|\Delta H_4| = |\Delta H_5|$

(2) $|\Delta H_2| > |\Delta H_1|$

(3) $|\Delta H_2| > |\Delta H_3|$

(4) $|\Delta H_1| = |\Delta H_6|$

12) EA of $O^+$ is more than :

(1) $N^+$

(2) $S^+$

(3) $F^+$

(4) $Ne^+$

13) Consider the hydrates ions of $Ti^{2+}$, $V^{2+}$, $Ti^{3+}$ and $Sc^{3+}$. The correct order of their spin-only magnetic moments is :

(1) $Sc^{3+} < Ti^{3+} < Ti^{2+} < V^{2+}$

(2) $Ti^{3+} < Ti^{2+} < Sc^{3+} < V^{2+}$

(3) $Sc^{3+} < Ti^{3+} < V^{2+} < Ti^{2+}$

(4) $V^{2+} < Ti^{2+} < Ti^{3+} < Sc^{3+}$

14)

| | **List-I (Order)** | | **List-II (Properties)** |
|---|---|---|---|
| (P) | K < Na < Li | (1) | Atomic radius |
| (Q) | Ca > Mg > Be | (2) | First ionisation energy |
| (R) | K > Ca | (3) | Metallic character |
| (S) | C > Si > Ge | (4) | Number of valence electrons |
| | | (5) | Second ionisation energy |

The correct code is :

(1) P→2,3; Q→3,4,5; R→1,2,4,5; S→2,5

(2) P→1,3,5; Q→2,4,5; R→1,5; S→2,4

(3) P→2,5; Q→1,3; R→1,3,5; S→2,5

(4) P→1,2; Q→1,3; R→1,3,5; S→1,2

15) The correct order of exchange energy :

(1) $p^3 < p^5 < p^6$

(2) $p^3 < p^6 < p^5$

(3) $p^5 < p^3 < p^6$

(4) $p^6 < p^5 < p^3$

16) Correct order of I.P of elements :-

(1) B > Tl > Al > Ga > In

(2) B > Tl > Ga > Al > In

(3) B > Al > Ga > In > Tl

(4) B > Tl > Ga > In > Al

17) Identify the wrong statement in the following:

(1) Atomic radius of the elements increases as one moves down the first group of the periodic table

(2) Atomic radius of the elements decreases as one-moves across from left to right in the 2nd period of the periodic table

(3) Amongst isoelectronic species, smaller the positive charge on the cation, smaller is the ionic radius

(4) Amongst isoelectronic species, greater the negative charge on the anion, larger is the ionic radius

18) If the formula of chloride of an element is $MCl_2$. Then the value of I.P. will be :-

| | $IP_1$ | $IP_2$ | $IP_3$ | $IP_4$ | $IP_5$ |
|---|---|---|---|---|---|
| (1) | 7.1 | 14.3 | 21.7 | 121.9 | 134.3 |
| (2) | 4.1 | 10.2 | 111.4 | 125.4 | 136.5 |
| (3) | 5.1 | 112.7 | 122.5 | 134.8 | 142.9 |
| (4) | 12.4 | 21.7 | 32.6 | 44.5 | 132 |

(1) 1

(2) 2

(3) 3

(4) 4

19) Which one of the following statement is **incorrect** about transition elements?

(1) The last electrons enters into them in the d-orbitals

(2) Their properties are in between those of s-block elements & p-block elements

(3) The transition elements with smallest atomic number is scandium

(4) Cn is a transition element

20) Identify the elements of third period whose successive I.E. are given below :-

| Element | $IE_1$ | $IE_2$ | $IE_3$(kJ/mol) |
|---|---|---|---|
| X | 1521 | 2666 | 3931 |
| Y | 513 | 4562 | 6920 |
| Z | 738 | 1451 | 7733 |

| | X | Y | Z |
|---|---|---|---|
| (1) | Na | Mg | Ar |
| (2) | C□ | S | Ar |
| (3) | Ar | Na | Mg |
| (4) | Mg | Ar | Na |

(1) 1

(2) 2

(3) 3

(4) 4

21) If $Z_{eff}$ of sodium is x then $Z_{eff}$ of Al is :-

(1) x + 0.90

(2) x + 1.30

(3) x + 0.65

(4) x

22) Which statement is **incorrect** :-

(1) Ionisation potential order : $F^- < Cl^-$

(2) Electron Affinity order : F < Cl

(3) Ionisation potential order : $O^- < S^-$

(4) Electronegatively order : $sp^3 > sp$ (for carbon atom)

23)

| **List-I** | | **List-II** | |
|---|---|---|---|
| **(Order)** | | **(Property)** | |
| (P) | $Na^+_{(g)} > Mg^{2+}_{(g)} > Al^{3+}_{(g)}$ | (1) | Size of ion/atom |
| (Q) | $F^-_{(g)} > O^-_{(g)}$ | (2) | Bond dissociation enthalpy |
| (R) | F > O > N | (3) | First ionisation enthalpy |
| (S) | $F_2 < O_2 < N_2$ | (4) | Electronegativity |

(1) P-1, Q-2, R-4, S-3

(2) P-4, Q-3, R-1, S-2

(3) P-1, Q-3, R-4, S-2

(4) P-2, Q-3, R-1, S-4

24) Which of the following is incorrect order for the indicated property ?

(1) Cr < Mn < Fe: Ionisation energy

(2) I < Br < F < Cl : Electron affinity

(3) $2^{nd}$ ionisation energy : C < N < F < O

(4) $N^- > P^- > O^- > S^-$ : Ionisation energy

25) Match the following.

| | **List-I** | | **List-II** |
|---|---|---|---|
| (P) | Atom with highest electronegativity | (1) | H |
| (Q) | Atom with highest electron affinity | (2) | F |
| (R) | Atom with lowest electronegativity | (3) | Cl |
| (S) | Atom with smallest radius | (4) | Cs |

(1) P→2; Q→4; R→1; S→3

(2) P→2; Q→3; R→4; S→1

(3) P→1; Q→4; R→2; S→3

(4) P→1; Q→2; R→3; S→4

26) In the compound M–O–H, the M–O bond will ionise in aq. medium readily if:

(1) Δ(E.N.) of M and O < Δ (E.N.) of O and H
(2) Δ(E.N.) of M and O = Δ (E.N.) of O and H
(3) Δ(E.N.) of M and O > Δ (E.N.) of O and H
(4) Cannot be predicated

27) The correct order of ionic radii of
$Y^{3+}$, $La^{3+}$, $Eu^{3+}$ and $Lu^{3+}$ is :-

(1) $Y^{3+} < La^{3+} < Eu^{3+} < Lu^{3+}$
(2) $Y^{3+} < Lu^{3+} < Eu^{3+} < La^{3+}$
(3) $Lu^{3+} < Eu^{3+} < La^{3+} < Y^{3+}$
(4) $La^{3+} < Eu^{3+} < Lu^{3+} < Y^{3+}$

28) Which of the following statement is **correct** ?

(1) Electron gain enthalpy may be positive for some elements
(2) Second electron gain enthalpy is always positive for all elements.
(3) $\Delta_{eg}H\ (K^{+}) = -\ IE\ (K)$
(4) All of these

29) The increasing order of electron affinity of the electronic configurations of element is :-
(I) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$
(II) $1s^2\ 2s^2\ 2p^3$
(III) $1s^2\ 2s^2\ 2p^5$
(IV) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^1$

(1) II < IV < III < I
(2) I < II < III < IV
(3) I < III < II < IV
(4) IV < III < II < I

30)

Which of the following relation is CORRECT according to Mulliken's Scale ?

(1) $X_M = \frac{I.P. + E.A.}{3}$
(2) $X_p = \frac{I.P. + E.A.}{2}$
(3) $X_p = \frac{I.P. + E.A.}{5.6}$
(4) $X_p = \frac{2.8}{X_M}$

31) Which orbitals are filled in given order in $5^{th}$ period of perodic table :-

(1) 5s → 4d → 5p

(2) 5s → 4d → 5d

(3) 4s → 5d → 4p

(4) 5s → 3d → 4p

32) In which of the following are the orders of ionisation energy of the elements or ions shown correctly ?

(1) $S < O^-$

(2) $O > S^-$

(3) $O^- > S^-$

(4) $O < S$

33) If electron gain enthalpies of Cl atom and $Cl^+$ ion are –348 KJ $mol^{-1}$ and –1251 KJ $mol^{-1}$ respectively, then the ΔH for the reaction $Cl^+_{(g)} \rightarrow Cl^-_{(g)}$ would be:

(1) –903 kJ

(2) –1599 kJ

(3) +903 kJ

(4) +1599 kJ

34) In the process $Cl(g) + e^- \xrightarrow{\Delta H} Cl^-(g)$, ΔH is

(1) Positive

(2) Negative

(3) Zero

(4) None

35) In which of the following pairs, atomic size of constituent elements are nearly the same or identical?
(A) Zr and Hf (B) Na and K
(C) Fe and Co (D) I and At
Choose the correct answer from the options given below :

(1) (A) and (B) only

(2) (B) and (C) only

(3) (A) and (C) only

(4) (C) and (D) only

36) Which of the following statement is **INCORRECT**?

(1) Li is strongest reducing agent among all elements in aqueous medium.

(2) Among $15^{th}$ group the $\Delta H_{eg}$ of N is slightly exothermic process

(3) The electron affinity of Cl is highest among all element.

(4) Radius of $Mg^{+2}$ is less than $Li^+$

37) Which is correct order of atomic size ?

(1) Al < Ge < Se
(2) Al < Se < Ge
(3) Ge > Al > Se
(4) Al > Ge > Se

38) Which of the following represent the correct order of atomic size.

(1) B < Al < Ga
(2) Sc < Y < La
(3) Ti > Hf > Zr
(4) Nd > Eu > Gd

39) The value of $IE_1$, $IE_2$, $IE_3$, and $IE_4$ of an atom are respectively 7.5 eV, 48.6 eV, 170.6 eV and 2200 eV. The electronic configuration of the atom will be :

(1) $1s^2\ 2s^2\ 2p^6\ 3s^1$
(2) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^1$
(3) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$
(4) $1s^2\ 2s^2\ 2p^6\ 3s^2$

40) Total number of elements which have more ionization energy as compare to their next higher atomic number elements. Li, Be, C, N, O, F, Ne

(1) 4
(2) 7
(3) 2
(4) 3

41) Match the following :-

| **List-I** | | **List-II** | |
|---|---|---|---|
| (P) | $3^{rd}$ transition series | (1) | IA |
| (Q) | Lanthanides | (2) | Electrons entered in (n–2) shell |
| (R) | Pnictogen | (3) | Half filled p-subshell |
| (S) | Most reactive metals | (4) | Au |

(1) P→4; Q→2; R→3; S→1
(2) P→4; Q→3; R→1; S→2
(3) P→3; Q→1; R→4; S→2
(4) P→4; Q→3; R→2; S→1

42) Match List-I with List-II and select correct code for your answer :-

| **List-I (Properties)** | | **List-II (Decreasing order)** | |
|---|---|---|---|
| (P) | Ionisation energy | (1) | $Ne > Na^+$ |
| (Q) | Atomic radius | (2) | S > O |
| (R) | Electronegativity | (3) | Hf > Zr |
| (S) | Electron gain enthalpy (Magnitude only) | (4) | O > N |

(1) P→1; Q→2; R→3; S→4

(2) P→1; Q→3; R→4; S→2

(3) P→4; Q→2; R→1; S→3

(4) P→3; Q→1; R→4; S→2

43) Which of the following set have correct order of Ionisation energy?

(1) Be < Mg < Ca

(2) A < Ga > In

(3) $F^- > Cl^-$

(4) H = $He^{\oplus}$

44) The property which is not common to both groups 1 and 17 elements in the periodic table are:

(1) Elelctropositive character increase down the groups.

(2) Reactivity decrease from top to bottom in these groups.

(3) Atomic radii increase as the atomic number increase.

(4) Electronegativity decrease on moving down a group.

45) The group in the periodic table that contains the elements in all the different physical states at room temperature is :

(1) V A

(2) I A

(3) VII A

(4) IV A

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ANSWER KEYS

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## CHEMISTRY

| Q. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A. | 4 | 3 | 3 | 2 | 4 | 3 | 4 | 4 | 3 | 1 | 3 | 2 | 1 | 3 | 1 | 2 | 3 | 2 | 4 | 3 |
| Q. | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| A. | 2 | 4 | 3 | 4 | 2 | 3 | 2 | 4 | 1 | 3 | 1 | 2 | 2 | 2 | 3 | 2 | 4 | 2 | 2 | 4 |
| Q. | 41 | 42 | 43 | 44 | 45 | | | | | | | | | | | | | | | |
| A. | 1 | 4 | 2 | 2 | 3 | | | | | | | | | | | | | | | |

# AIM: FREE EDUCATION TO ALL APUL

SOLUTIONS

CHEMISTRY

1) atom can't be 3rd period element as 3rd period element can't have 3d-orbitals in it's penultimate shell

2)

| | Period and group number respectively |
|---|---|
| (P) $_{14}Si \Rightarrow [Ne]\ 3s^2 3p^2$ | → 3, 14 |
| (Q) $_{21}Sc \Rightarrow [Ar]\ 3d^1 4s^2$ | → 4, 3 |
| (R) $_{31}Ga \Rightarrow [Ar]\ 3d^{10} 4s^2 4p^1$ | → 4, 13 |
| (S) $_{73}Tl \Rightarrow [Xe]\ 4f^{14} 5d^{10} 6s^2 6p^1$ | → 6, 13 |

3)

Size : $F^- < Cl^- < Br^- < I^-$

4) **New Solutions :-**

For s and p block element, $Z_{eff}$ increases by 0.65 on moving left to right

N O

$Z_{eff}$ Y Y + 0.65

For N : $Z_{eff} = Z - \sigma$

$Y = 7 - \sigma$

$\sigma = 7 - Y$

For 0 : $Z_{eff} = Z - \sigma$

$Y + 0.65 = 8 - \sigma$

$\sigma = 7.35 - Y$

**Old Solutions :-**

Explanation : σ of oxygen is asked w.r.t. Nitrogen

**Concept :** Leater mal

**Solution/Explanation/Calculation :**

$Z_{eff}$. of N = Y is given

$Z_{eff} = Z - \sigma$

$= 7 - 3.10$

$= 3.90 = Y$

'O' (z = 8) = $1s^2 2s^2 2p^4$

N = $1s^2\ 2s^2\ 2p^3$

$s = 4 \times 0.35 + 2 \times 0.85$

$= 1.40 + 1.70$

$= 3.10$

Oxygen has $1e^-$ alone than Nitrogen thus it's Zeff is just Most than 0.65 of Nitrogn
 s of 'O' = 7.35 – Y
Hence, the correct answer is option (2).

5)

$Z_{eff}$ is same for (ns, np) electron .

6) $Na^+$ and $F^-$ isoelectronic but Na belongs to $3^{rd}$ period while F belongs to $2^{nd}$ period.

7)

| Atomic Number | Ionization Enthalpy (kJ/mol) | | |
|---|---|---|---|
| | $I_1$ | $I_2$ | $I_3$ |
| n | 1579 | 3374 | 6043 |
| n+1 | 2075 | 3952 | 6125 |
| n+2 | 493 | 4562 | 6907 |
| n+3 | 739 | 1451 | 7731 |

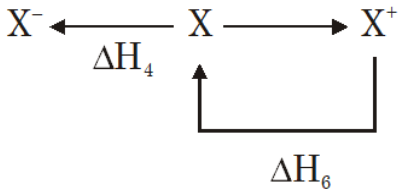
By observing the values of $I_1$, $I_2$ & $I_3$ for atomic number (n + 2), it is observed that $I_2 >> I_1$.
This indicates that number of valence shell electrons is 1 and atomic number (n+2) should be an alkali metal.
Also for atomic number (n + 3), $I_3 >> I_2$.
This indicates that it will be an alkaline earth metal which suggests that atomic number (n + 1) should be a noble gas & atomic number (n) should belong to Halogen family. Since n < 10; hence n = 9 (F atom)
Note: n = 1 (H atom) cannot be the answer because it does not have $I_2$ & $I_3$ values.

8) Zn has fully-filled orbitals. As a result, it has higher IE than Ga.

9) $\Delta E_{eg}$ order
(F) < ($F^+$) {positive charge so
(Cl) < ($Cl^+$) more attraction}
&
|I.E. of F| = |$\Delta H_{eg}$ of $F^+$|
|I.E. of Cl| = |$\Delta H_{eg}$ of $Cl^+$|
I.E. order ⇒ F > Cl
So $\Delta H_{eg}$ order ⇒ $F^+ > Cl^+$ and
order of $\Delta H_{eg}$ ⇒ $F^+ > Cl^+ > Cl > F$
↓ ↓ ↓ ↓
(E.A. value ⇒ 17.4 13 3.6 3.4
in eV/atom)
Ans. (C) P→1; Q→2; R→4, S→3

10)

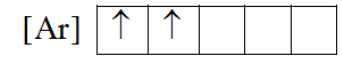
$N \rightarrow N^-$
$2p^3$




11)
$\Delta H_3 > \Delta H_2$
$IE_3 > IE_2 > IE_1$

12) IE of A = EA of $A^+$

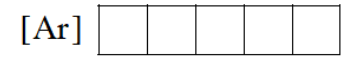
13) Spin only magnetic moment = $\sqrt{n(n+2)}$ B.M.
Electronic configuration of
$Ti^{2+} \Rightarrow [Ar]\ 3d^2\ 4s^\circ$



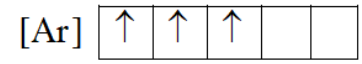

Magnetic Moment (m) = $\sqrt{8}$B.M.
$Ti^{3+} \Rightarrow [Ar]\ 3d^1\ 4s^\circ$




Magnetic Moment (m) = $\sqrt{3}$B.M.
$Sc^{3+} \Rightarrow [Ar]\ 3d^\circ\ 4s^\circ$




Magnetic Moment (m) = 0 B.M.
$V^{2+} \Rightarrow [Ar]\ 3d^3\ 4s^\circ$




Magnetic Moment (m) = $\sqrt{15}$ B.M.
So order of spin only magnetic moment is :
$Sc^{3+} < Ti^{3+} < Ti^{2+} < V^{2+}$

14) Theory base.

15) Stability ∝ Exchange energy

16) Value bases
B > Tl > Ga > Al > In

17) At. radii $\propto \frac{\text{–ive charge}}{\text{+ive charge}}$

18) Valency of M = 2 in 2nd option sudden jump occurs after 2nd IP.

19) Lawrencium is f - block element.

20) Inert gas have max. I.P & Alkali metal has min I.P.

21) $\underset{x}{Na}$ $\underset{x+0.65}{Mg}$ $\underset{x+0.65+0.65}{Al}$
In 3rd period $Z_{eff}$ increases by 0.65

22) EN ∝ % s-character sp > $sp^2$ > $sp^3$

23) (P) $Na^+ > Mg^{2+} > Al^{3+}$ (Size of ion)
(Q) $F^- > O^-$ (First ionisation enthalpy)
(R) F > O > N (Electronegativity)
(S) $F_2 < O_2 < N_2$ (Bond dissociation enthalpy)

24)

(A) For d-block elements :
In a period very irregular pattern is observed by for these '3' consecutive elements.
Cr < Mn < Fe (IE order)

(B) Electron affinity :
3rd period element > 2nd period elements
(due to larger size of 3rd period elements
(if both elements in same group)
In 17th group
F, Cl, Br, I
$\underbrace{Cl}_{\text{Due to larger size of 'Cl', EA more}} > \underbrace{F > Br > I}_{\text{Down the group EA ↓es}}$

(C) After 1st ionisation :
Electronic configuration :
$C^+ \rightarrow 1s^2\ 2s^2\ 2p^1$
$N^+ \rightarrow 1s^2\ 2s^2\ 2p^2$
$O^+ \rightarrow 1s^2\ 2s^2\ 2p^3$ Half filled, so IE ($O^+$) more
$F^+ \rightarrow 1s^2\ 2s^2\ 2p^4$
1st IE order :
$O^+ > F^+ > N^+ > C^+$
2nd IE order
O > F > N > C
$\left[M^+ \xrightarrow{2^{nd}\ IE} M^{+2} + e^-\right]$

(D) $N \xrightarrow{+e^-} N^-$
$P \xrightarrow{+e^-} P^-$
$O^- \xrightarrow{+e^-} O^-$
$S \xrightarrow{+e^-} S^-$
Electron affinity order
S > O (3rd period element > 2nd period element)
S > P (in a group left to right EA increases)

$N^- \rightarrow N + e^-$
$P^- \rightarrow P + e^-$
$S^- \rightarrow S + e^-$
$O^- \rightarrow O + e^-$
} I.E.

I.E. order :

$S^- > O^-$ (If EA more for neutral species $S^- > P^-$ then IE more for its anions.)

D ⇒ incorrect order.

25) Theory based.

26) With increasing in electronegative difference ionisation occur radily.

27) $Y^{3+} < La^{3+}$
$Lu^{3+} < Eu^{3+} < La^{3+}$ } (Order of radii)

28) From notes

29) N < Al < F < Cl

30)

Correct relation according to mulliken' scale-

(C) $x_p = \frac{I.P. + E.A}{5.6}$

31) According to n + l rule electron will be filled in the following order of orbitals 5s → 4d → 5p

32) EA of neutral atom is generally higher than anion.

33)
$$Cl^+ + e^- \longrightarrow Cl; \quad \Delta H = -348\,kJ$$
$$Cl + e^- \longrightarrow Cl^- \; ; \quad \Delta H = -1251\,kJ$$
$$Cl^+ + 2e^- \longrightarrow Cl^- ; \Delta H = -1599\,kJ$$

34)

Ans is (B)

$Cl_{(g)} + e^{\ominus} \rightarrow Cl^{\ominus}_{(g)}$

ΔHe is negative

35) Rb & Cs have nearly same electron gain enthalpy electron gain enthalpy = –46 kj/ml
Ar & Kr have same $\Delta H_{eq}$. Value is + 96 kj/ml.

36)

Which of the following statement is Incorrect?
(B) Among 15$^{th}$ group the $\Delta H_{eg}$ of N is sligtly exothermic process.

⇓

Incorrect

$N + e^- \rightarrow N^-$ $\Delta H_{eg}$ = (+)ve
endothermic process
because $\simeq$ $\Delta H_{eg}$ –E.A.
E.A of N = Negative
So $\Delta H_{eg}$ = Positive

37) Al 125 Pm
Ge 122 Pm
Se 114 Pm

38) In d–block, in group 3, atomic size increases down the group.

39) Differnce in 3$^{rd}$ and 4$^{th}$ ionization energy is high so only three electron can be removed easily

40)

Li < Be
Be > B
C < N
N > O
O < F
F < Ne
Ne > Na

} Ionization energy order

41)
[P] 3rd transition series → [4] Au [d-block element]
[Q] Lanthanides → [2] e–entered in (n-2),(n–1), n series
[R] Pnictogen → [3] [V] A series known as pnitogen so half filled p-orbital.
[S] Most Reactive metals → [1] IA series is most reactive due to unpaired $e^-$ in outer most shell.

42) [P] Ionisation energy → [3] HF > Zr [I.E.]
[Q] Radii → [1] Ne > $Na^+$

[R] Electronegativity → O > N
[S] Electron gain enthalpy → [2] S > O
↓
[More space]

43)

Isoelectronic species have correct order of ionisation energy-

$$\text{Atomic size} : IE \propto \frac{1}{\text{Atomic size}}$$

(B) $N < O^{\oplus}$ is the correct answer.

44) Down the group reactivity increases for Ist group but decreases for 17th group.

45) F gas
Cl gas
Br liquid
I Solid